🚀 MAJOR PROJECT 7 (FINAL):

SECURE FILE TRANSFER USING ZENITY & SCP (AWS EC2)

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1. Introduction

In modern computing environments, secure file transfer is a critical requirement, especially when working with remote systems and cloud infrastructure. This project demonstrates a secure and automated file transfer system using **Zenity (GUI)** and **SCP (Secure Copy Protocol)**.

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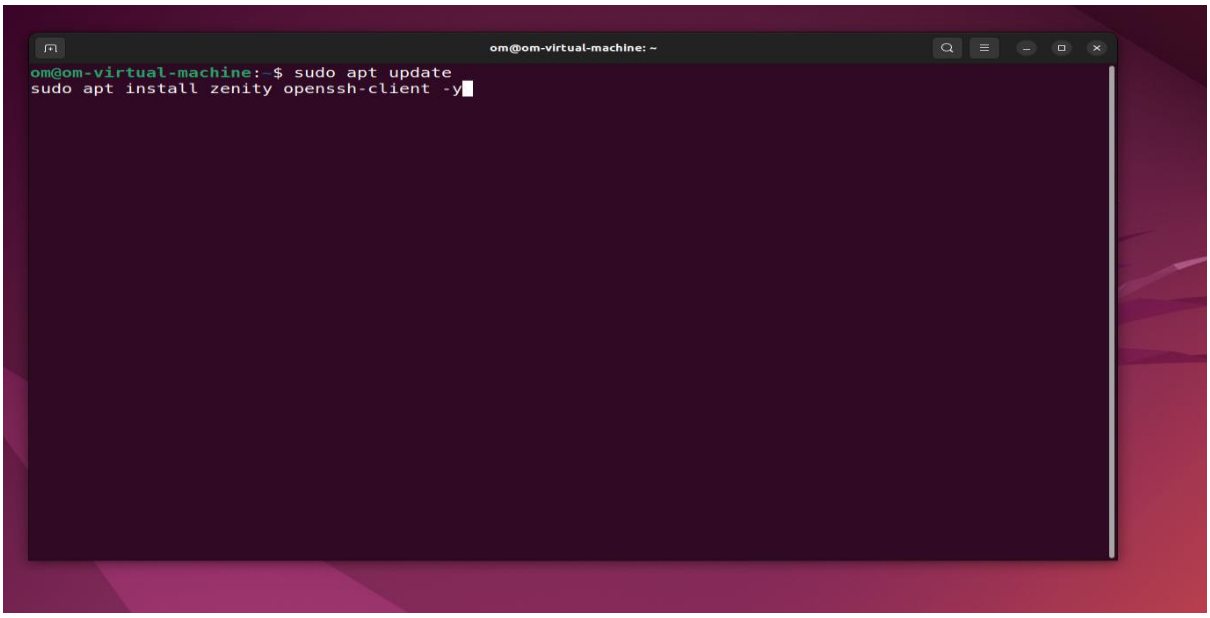
2. Objective

- To develop a GUI-based secure file transfer system
- To implement file transfer using SCP over SSH
- To automate remote file transfer using shell scripting
- To understand cloud-based communication using AWS EC2

🔧 3. Prerequisites

On Local VM:

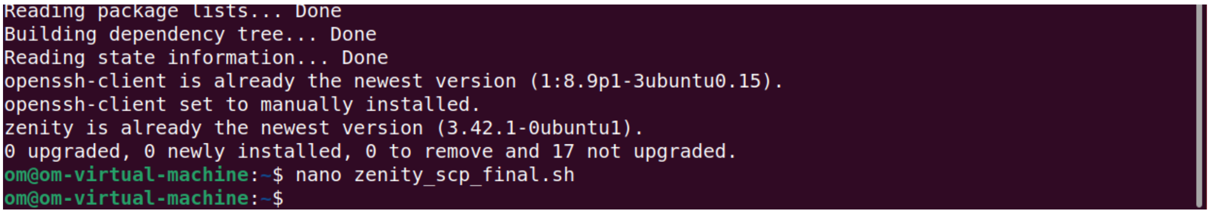
Command: sudo apt update
sudo apt install zenity openssh-client -y



💻 4. Final Script (Production Version)

Create file:

Command : nano zenity_scp_final.sh



```
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
openssh-client is already the newest version (1:8.9p1-3ubuntu0.15).
openssh-client set to manually installed.
zenity is already the newest version (3.42.1-0ubuntu1).
0 upgraded, 0 newly installed, 0 to remove and 17 not upgraded.
om@om-virtual-machine:~$ nano zenity_scp_final.sh
om@om-virtual-machine:~$
```

Paste this 👇

```
#!/bin/bash

TITLE="Secure File Transfer (Zenity + SCP + AWS)"

# Step 1: Select File
FILE=$(zenity --file-selection --title="Select File to Transfer")

if [ -z "$FILE" ]; then
   zenity --error --text="No file selected!"
   exit 1
fi

# Step 2: Enter EC2 Public IP
IP=$(zenity --entry --title="$TITLE" --text="Enter EC2 Public IP")

if [ -z "$IP" ]; then
   zenity --error --text="IP Address required!"
   exit 1
fi

# Step 3: Enter Username (default: ubuntu)
USER=$(zenity --entry --title="$TITLE" --text="Enter Username (e.g. ubuntu)" --entry-
text="ubuntu")

if [ -z "$USER" ]; then
   zenity --error --text="Username required!"
   exit 1
fi

# Step 4: Select PEM Key
KEY=$(zenity --file-selection --title="Select PEM Key File")

if [ -z "$KEY" ]; then
   zenity --error --text="PEM key required!"
   exit 1
fi

# Fix permission automatically
chmod 400 "$KEY"
```

# Step 5: Destination Path
DEST=$(zenity --entry --title="$TITLE" --text="Enter Destination Path" --entry-text="/home/ubuntu/")

if [-z "$DEST"]; then
  zenity --error --text="Destination required!"
  exit 1
fi

Step 6: Transfer with Progress
(
echo "10"; echo "# Connecting to EC2..."
sleep 1

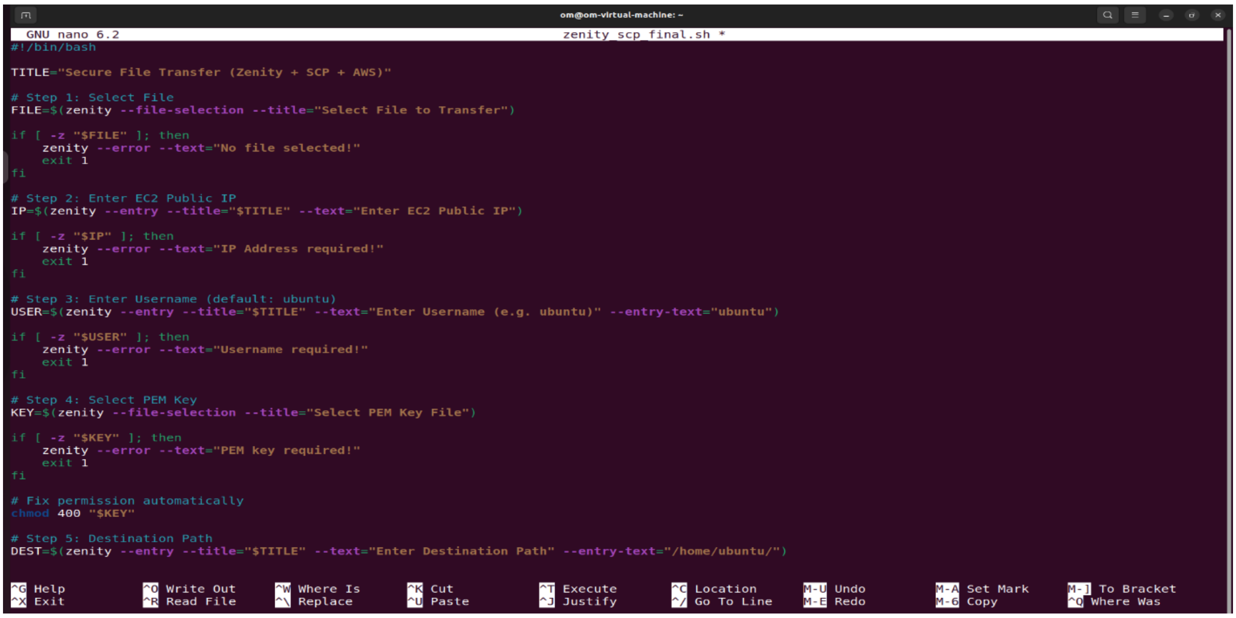
echo "40"; echo "# Uploading file..."

scp -i "$KEY" -o StrictHostKeyChecking=no "$FILE" "$USER@$IP:$DEST"

if [$? -eq 0]; then
  echo "100"; echo "# Transfer Completed!"
else
  echo "100"; echo "# Transfer Failed!"
  exit 1
fi

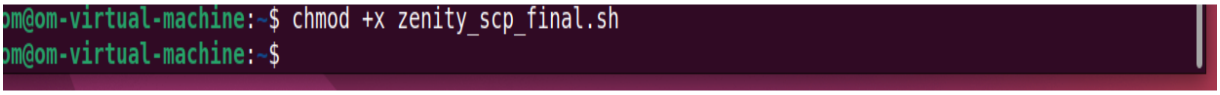
) | zenity --progress \
--title="$TITLE" \
--percentage=0 \
--auto-close

# Final Status
if [$? -eq 0]; then
  zenity --info --text="✅ File transferred successfully to EC2!"
else
  zenity --error --text="❌ Transfer failed!"
fi



🔐 5. Make Executable

Command : chmod +x zenity_scp_final.sh



▶️ 6. Run Project

Command : ./zenity_scp_final.sh

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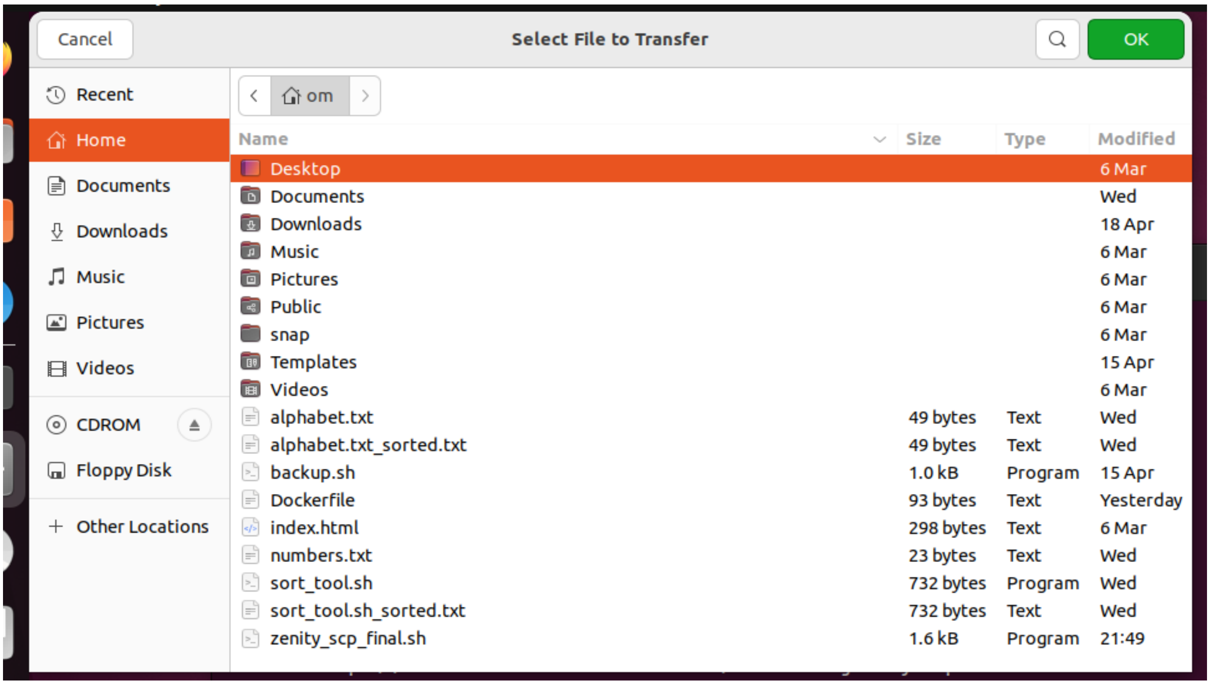
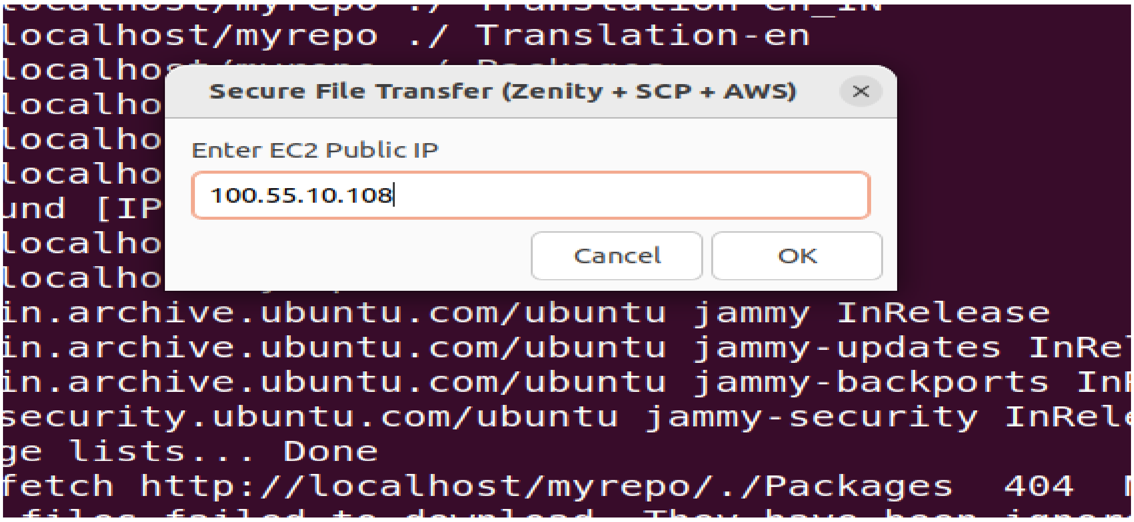
7. Workflow (Write in Report)

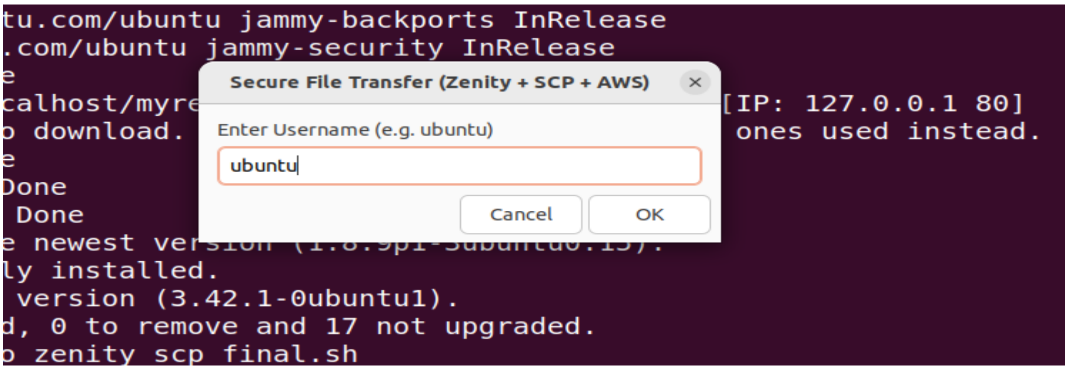
1. User selects file using Zenity GUI



2. User enters EC2 Public IP

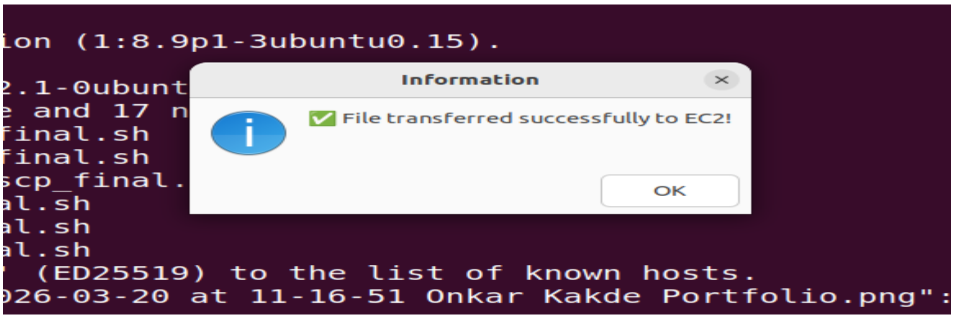


3. User enters username (ubuntu)

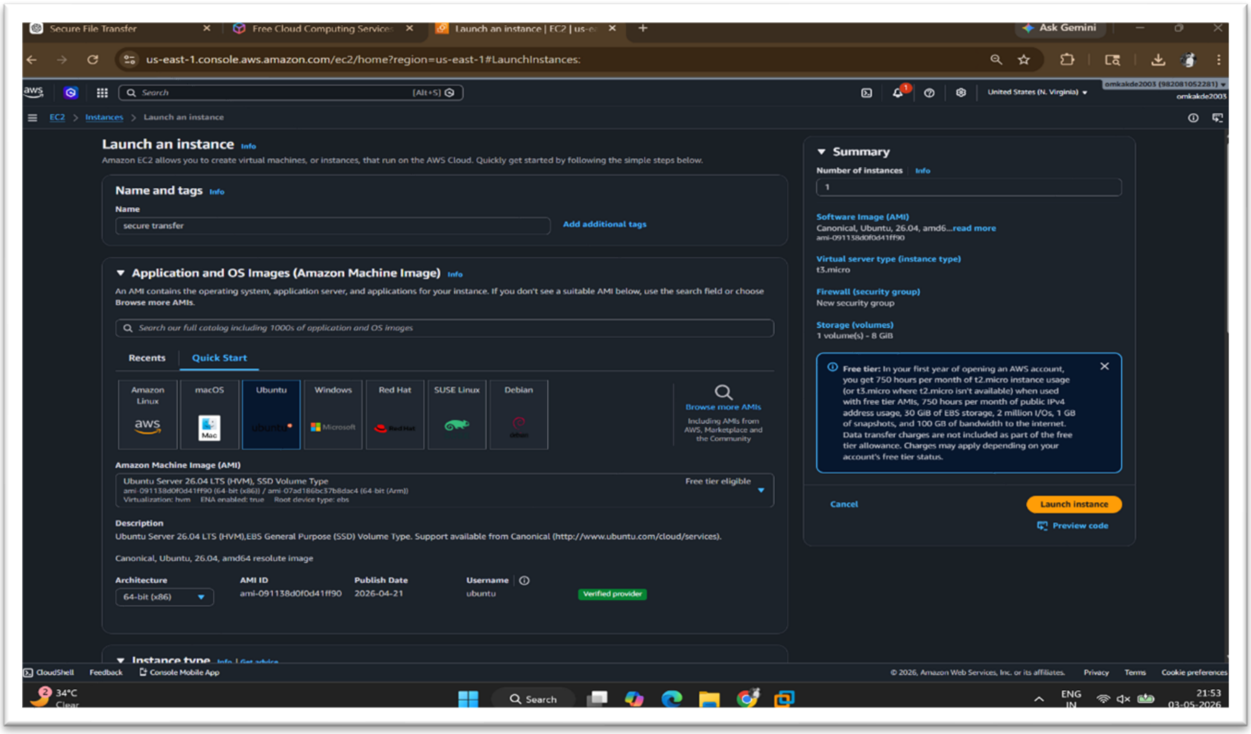


4. User selects PEM key file

5. Success/error message displayed



Aws console :



Architecture for project